

Upcoming Webinar

How should we be performing time series simulation studies?

DATE

Monday, September 28, 2026

TIME

12:00 PM - 1:30 PM ET

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Abstract

As a community, we don't do simulation studies well - myself included! We often produce theory for general ARMA orders but give simulation studies with one anecdotal case or maybe a grid of values for an AR or MA order 1 process. Given the wide applicability and use of these models in the wild, we should be demonstrating when methods work well and when they don't in an honest way so practitioners can choose the best methods for the data at hand.

To run robust simulations involving ARMA (Autoregressive Moving Average) generated data rather than picking anecdotal cases, it is preferable to choose parameters representative of the full sample space corresponding to causal and invertible difference equations. Specific parameter values can either be selected via a fixed design experiment or a random effects experiment. Implementation of either of these methods requires quantifying these parameter spaces. These spaces are not described for ARMA orders higher than two in time series texts, nor are they readily available in the literature. This talk describes how to determine the parameter spaces for higher-order processes, with explicit descriptions and graphics of the parameter space up to order 4. To randomly generate parameters in these spaces, methods that generate parameters and use roots of polynomials to check for causality are highly inefficient, while first generating roots of polynomials and then determining parameters is cumbersome and lacks a simple connection to the parameter space. We provide an efficient algorithm to generate parameters within these spaces. Parameters can be generated uniformly across possible correlations or uniformly in the parameter space. Furthermore, a simple measure of total correlation is proposed for autoregressive processes of arbitrary order, which can be used within simulation studies to test statistical methods under "high" and "low" correlation scenarios.